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CS110 Lab 4

Github Link: https://github.com/Drew-Kim/CS110Lab4

Part 1 - New York Times API

For Part 1, I created a separate Part1 folder that contains the required files for a New York Times Popular Articles page that lets the user choose:

- Popularity criteria: Most Viewed, Most Shared, or Most Emailed
- Time frame: Day, Week, or Month

The HTML was placed in index.html, the CSS was placed in style.css, and the JavaScript was placed in main.js, following the lab directions.

- index.html
- style.css
- main.js
- README.md

Fetch API

The project uses the JavaScript Fetch API in main.js to make a GET request to the NYT Popular Articles API. The request URL changes depending on the selected popularity criteria and time frame.

Article Display

The page displays up to 5 popular articles. Each article item includes:

- An image
- A title
- An abstract/description
- A published date

Each article is shown inside its own separate card, so the articles are easy to read.

Responsive Design

The CSS uses media queries so the layout changes on smaller screens.

On wider screens, the filter box appears on the left, and the article list appears on the right. On screens smaller than 900px, the page changes to a single-column layout with the filter box above the articles.

Documentation

I added JSDoc-style function comments in main.js for the main JavaScript functions. I also added a README.md file explaining how the API key is used and noting that no external libraries were used for Part 1.

Part 2 - React Tic Tac Toe

For Part 2, I created a separate Part2 folder with a React project.

Important files include:

- src/Board.jsx
- src/App.jsx
- src/RulesPage.jsx
- src/main.jsx
- src/styles.css
- package.json
- README.md

Main Game Features:

- X always starts the game.
- The app keeps track of whose turn it is.
- The app keeps score for X and O.
- The app calculates and displays the winner.
- The app includes a New Game button.

Player Turn Tracking

The game uses React state to track whether X or O should play next.
When a player clicks an empty square, the board updates and the turn switches to the other player.

Score Tracking

The game uses React state to store the scores for X and O.
When a player wins, that player's score increases by 1. The New Game button clears the board but keeps the current score.

Winner Calculation

If one player has three matching marks in one of those lines, the game displays that player as the winner. The calculateWinner function checks all possible winning combinations:

- Three rows
- Three columns
- Two diagonals

New Game Button

The New Game button resets the board to an empty Tic Tac Toe board and makes X the first player again. The score does not reset when starting a new game, so users can play multiple rounds.

Extra Credit - React Routing

For the extra credit, I added React Routing to Part 2 using the react-router-dom library.
The app now has two pages:

- / - Tic Tac Toe game page
- /rules - Rules page

The new page is implemented in src/RulesPage.jsx.
I also added a navigation bar in App.jsx so the user can move between the Game page and the Rules page. The README.md file for Part 2 was updated to mention that react-router-dom is used for routing.

Summary

Part 1:

- No external libraries were used.

Part 2:

- React was used to build the Tic Tac Toe game.
- react-router-dom was used for the extra credit React Routing requirement.

Testing and Verification

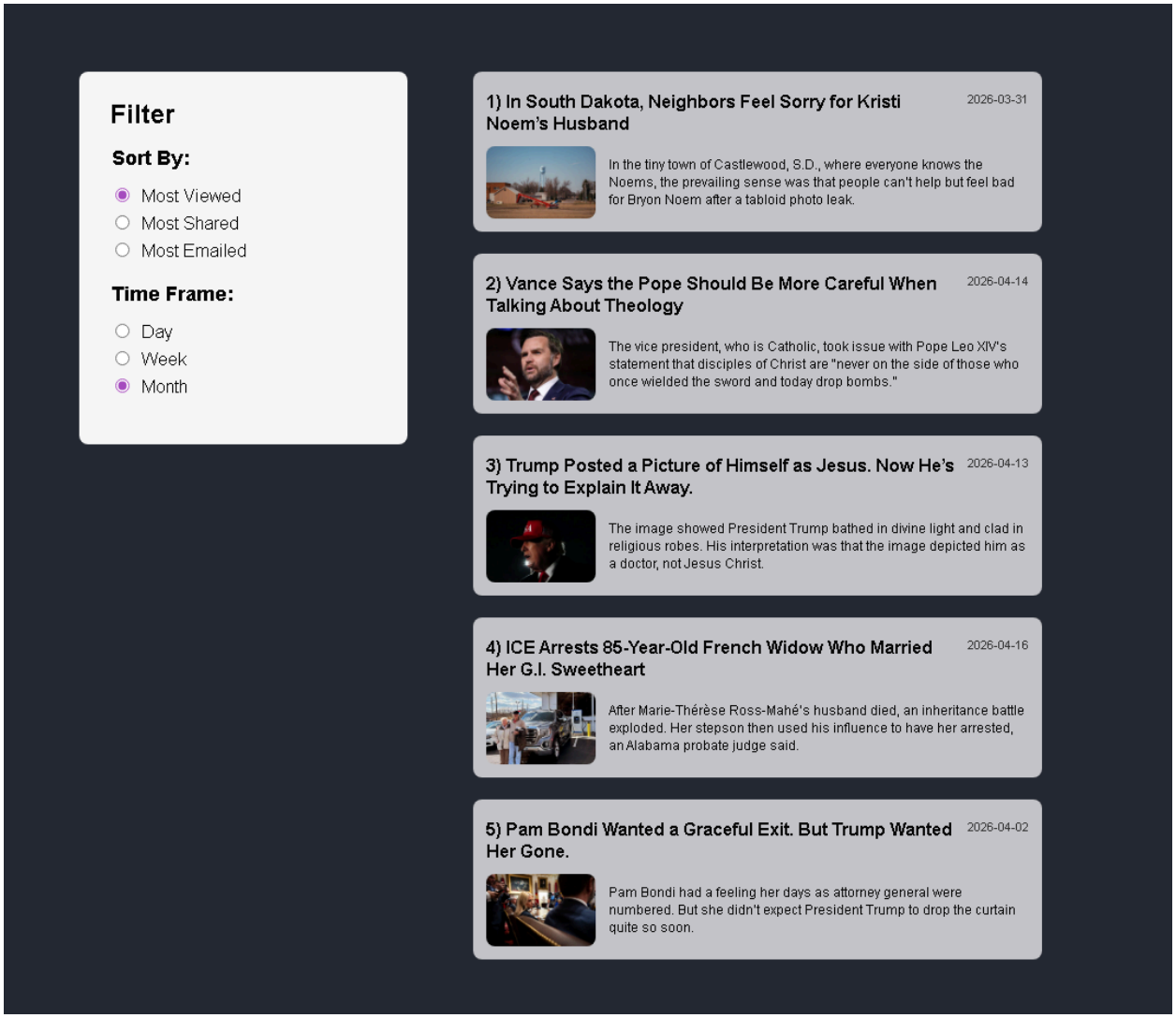
Part 1:

- The JavaScript file was checked with node --check.
- The NYT API key was added to main.js.

Part 2:

- Dependencies were installed with npm install.
- The React app was built successfully with npm run build.
- The local development server was tested.
- The game route / and rules route / rules both loaded successfully.

Part 1: Demo



Responsive (>900)

Filter
Sort By:
Most Viewed
Most Shared
Most Emailed
Time Frame:
Day
Week
Month
1) In South Dakota, Neighbors Feel Sorry for Kristi Noem's Husband
2026-03-31
In the tiny town of Castlewood, S.D., where everyone knows the Noems, the prevailing sense was that people can't help but feel bad for Bryon Noem after a tabloid photo leak.
2) Vance Says the Pope Should Be More Careful When Talking About Theology
2026-04-14
The vice president, who is Catholic, took issue with Pope Leo XIV's statement that disciples of Christ are "never on the side of those who once wielded the sword and today drop bombs."
3) Trump Posted a Picture of Himself as Jesus. Now He's Trying to Explain It Away.
2026-04-13
The image showed President Trump bathed in divine light and clad in religious robes. His interpretation was that the image depicted him as a doctor, not Jesus Christ.

Part 2: Demo

Extra Credit: Rule Page

Game Rules

How to Play

X always goes first. Players take turns choosing an empty square until someone wins or the board is full.

- Get three marks in a row to win.
- Rows, columns, and diagonals all count.
- The New Game button clears the board but keeps the score.

Back to Game